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I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references. Place, date and signature.

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Plugin System's in Rust

Gabriel Zimmermann

I. Abstract

Rust is a systems programming language that has gained significant popularity in recent years due to its focus on safety, performance, and concurrency, but the extendability is currently limited. In this thesis, we explore the advantages and disadvantages for a selection of Rust crates when designing a dynamic (does plugin system already imply it's dynamic?) plugin system.

Contents

I)	Abstract	3
II)	Introduction	6
II.A)	Motivation	6
II.A.a)	Plugin Systems need dynamic linking	6
II.A.b)	Staying true to Rust's features	6
II.B)	Research Questions	6
III)	Related work	7
III.A)	Definitions	7
III.A.a)	Plugin System	7
III.A.b)	ABI	7
IV)	Methods	8
IV.A)	Rust API	8
IV.A.a)	Unstable Rust ABI	8
IV.A.b)	Stable Rust ABI (crate)	8
IV.B)	C ABI	8
IV.B.a)	"Raw" usage	8
IV.B.b)	Crate "stabby"?	8
IV.B.c)	Other crates (although no focus here)	8
IV.C)	IPC	8
IV.C.a)	Crate "rust_bridge" (googling this is horrible qwq)	8
IV.C.b)	Other crates	8
V)	Results	9
V.A)	Conclusion and Future Work	10
VI)	Listings	11
VII)	Code Snippet	12
	References	12

II. Introduction

A. Motivation

a) Plugin Systems need dynamic linking:

Rust does not have a stable ABI, which means that the compiler can make changes to the way it generates code that can break compatibility with existing binaries. This makes it difficult to create plugins that can be loaded at runtime, as the plugin and the host application may not be compatible.

b) Staying true to Rust's features:

When leaving the Rust ecosystem (like using C Abi), you lose some of the features that Rust provides, such as memory safety and type safety. This can make it more difficult to write plugins that are safe and reliable.

B. Research Questions

Understanding the advantages and disadvantages of different approaches to designing a plugin system in Rust can help developers make informed decisions about how to implement their own plugin systems. This research can also contribute to the broader understanding of how to design extensible systems in Rust, which can benefit the Rust community as a whole. (Okay that last sentence just screams AI :))

III. Related work

A. Definitions

a) *Plugin System*:

b) *ABI*:

No comprehensive comparison between possible solutions for plugin systems in Rust has been done yet. There are some simple comparisons, but they don't cover the full range of options and trade-offs.

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IV. Methods

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A. Rust API

- a) *Unstable Rust ABI:*
- b) *Stable Rust ABI (crate):*

B. C ABI

- a) *"Raw" usage:*
- b) *Crate "stabby"?):*
- c) *Other crates (although no focus here):*

C. IPC

- a) *Crate "rust_bridge" (googling this is horrible qwq):*
- b) *Other crates:*

V. Results

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A. Conclusion and Future Work

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VI. Listings

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VII. Code Snippet

```
fn main() {  
    println!("Hello world!")  
}
```

Listing 1. Rust Code

References